

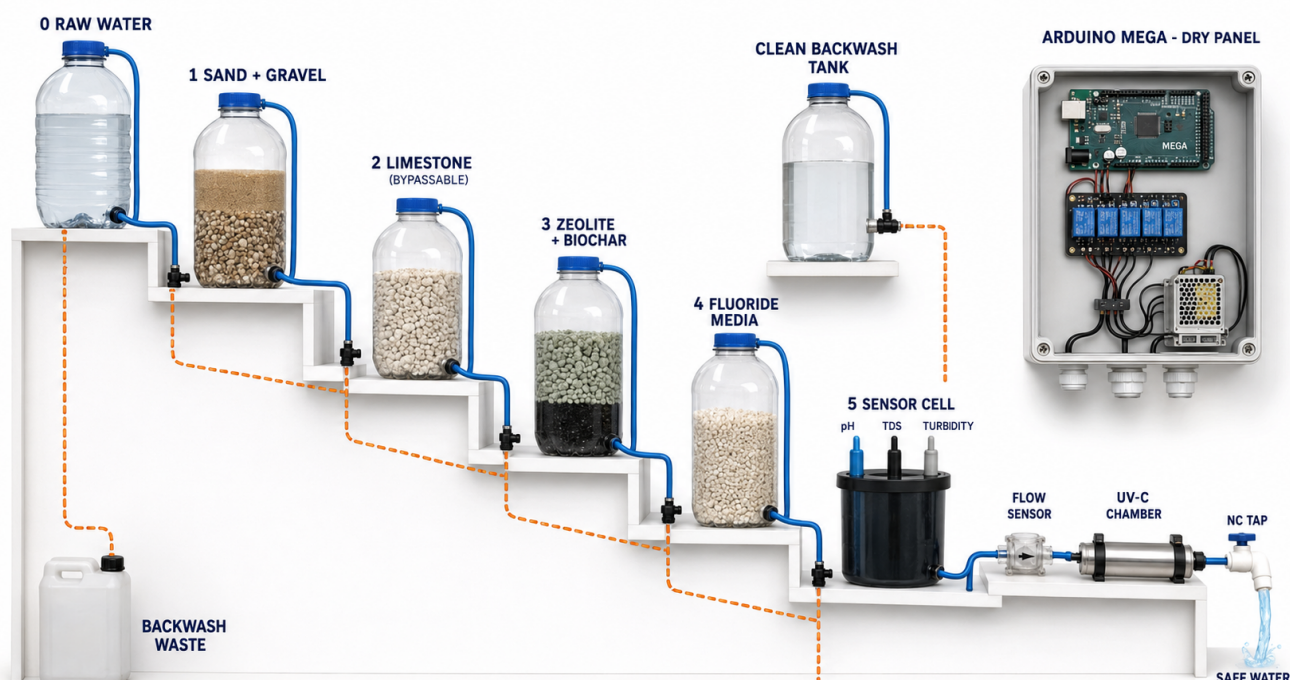
ProtoWave

Arduino Mega gravity-fed water purification and quality interlock

Sensor placement, stepped-bottle hydraulics, evidence-based thresholds, safe wiring, firmware and validation plan

DESIGN POSITION

Build a **descending, modular treatment train**. Use the listed sensors for process control, not as proof that fluoride or heavy metals are absent. Keep the drinking-water tap fail-closed until calibration and validation are complete.



Prototype for Smart India Hackathon 2026 problem statement SIH26040

Design revision: Arduino Mega 2560 | Fail-closed control | Gravity service and separate reverse-flow backwash

1. Executive engineering decision

Decision	Recommended implementation	Reason
Controller	Arduino Mega 2560	16 analog inputs and four hardware serial ports leave enough I/O for sensors, valves and SIM800L without software serial.
Hydraulics	Descending bottle staircase	Preserves available static head, avoids uphill loops and reduces trapped air at low pressure.
Treatment	Source-specific, replaceable cartridges	High-fluoride groundwater and acid mine drainage require different chemistry; limestone must be bypassable.
Sensors	One final online sensor cell before UV	Measures the quality actually presented to the disinfection barrier and tap.
Safety	Normally-closed tap plus software interlocks	Loss of power, invalid calibration or unsafe measurements closes the drinking-water path.
Fluoride	Optional ISE/colorimetric side-stream	pH/TDS/turbidity do not directly measure fluoride. Reagent-conditioned sample must drain to waste.
Metals	Accredited lab release testing; optional validated ASV R&D; module	The listed sensor set cannot identify As, Pb, Cd, Cr, Ni, Hg, Mn, Cu or Fe.

Blunt corrections to the original concept

- A single motorized valve cannot reverse flow. True gravity backwash needs separate feed, tap, clean-backwash and waste paths, or a pump.
- A generic 12 V UV-C LED chamber is not automatically a disinfection barrier. Delivered dose, water transmittance, flow and lamp/intensity status must be validated.
- Automatic tap locking is defensible only for parameters that are actually measured and for sensor failures. Do not label a TDS excursion as fluoride or metal toxicity.
- Transparent PET is useful for a short judging demonstration. A deployable unit needs opaque, cleanable, UV-stable and pressure-rated housings.

The award-winning version is the one that clearly separates what is measured, what is inferred, and what is laboratory-verified.

2. Why the filters should step downward

Yes - the bottle prototype should descend from the raw-water reservoir to the final tap. The staircase is not a series of pumps and it does not multiply pressure. It protects the single pressure source already available: elevation.

$$P = \rho g h$$

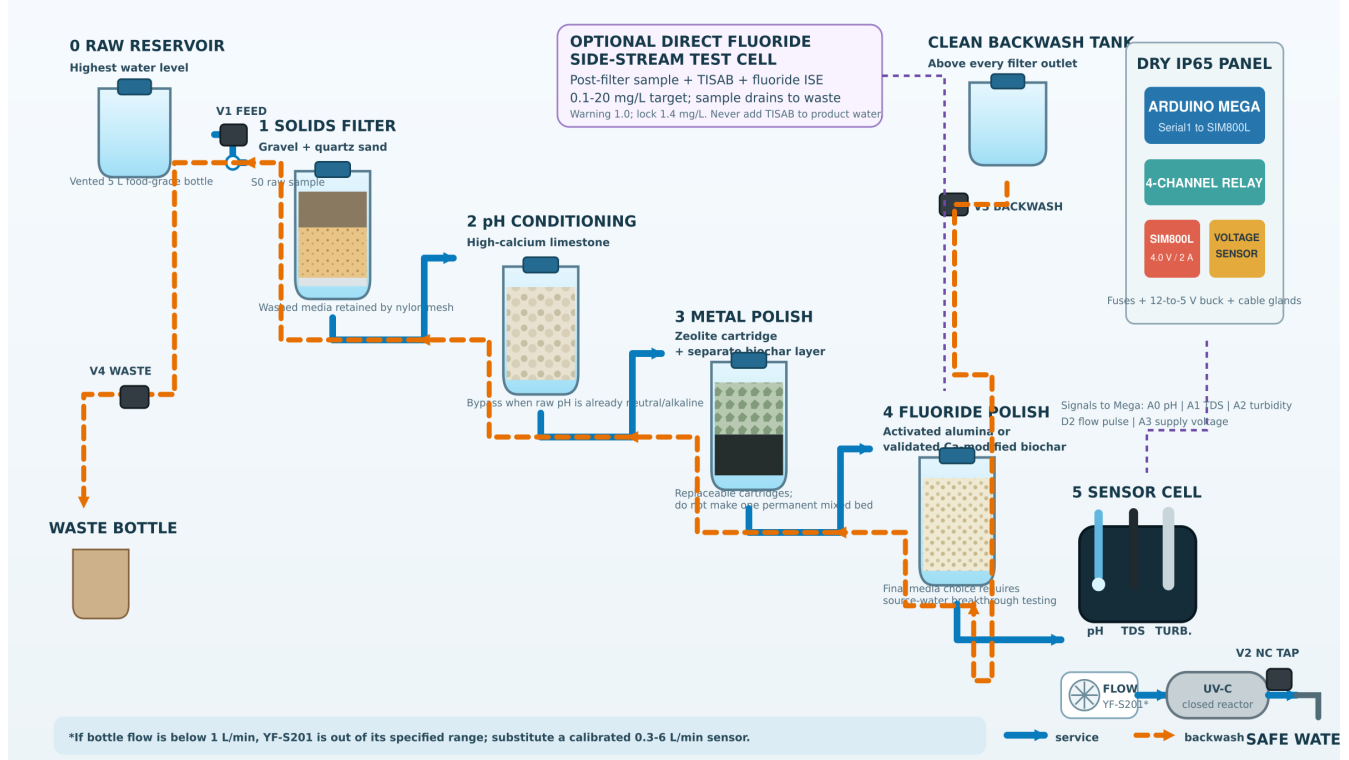
For water: 1.0 m head is approximately 9.8 kPa or 0.098 bar.

Layout	Will it flow?	Engineering consequence
Descending staircase	Best passive option	Every inter-stage connection can stay level or fall. Venting is simple; bubbles rise out rather than collecting at high points.
Same-height row	Can work	Only if the raw surface remains above the tap, tubing is fully primed and pressure losses stay below the available head.
Tubing rises between stages	Unreliable at bottle scale	Creates high points that trap air and consume head; flow may stop as media loads.

- Set the raw-water surface at least 1.0-1.5 m above the tap for the prototype; confirm by measurement rather than relying on the drawing.
- Give each bottle a vented top inlet and a bottom outlet. Use short, large-bore food-grade tubing and avoid sharp elbows.
- A 100-150 mm downward step between bottle shelves is practical. Total raw-surface-to-tap height matters more than identical step height.
- Measure flow at high and low tank levels, then with partly clogged media. The lowest measured head is the design case.
- Backwash is different: place a clean-water tank above every filter and send clean water upward through the selected bed to a separate waste bottle.

ProtoWave bottle prototype - equipment and sensor placement

Normal service flow descends left-to-right; backwash uses a separate elevated clean-water head.



Exact equipment and sensor placement. Blue is service flow; orange dashed is the separate backwash path.

3. Bottle train and construction

Stage	Prototype construction	Purpose and design caution
0 Raw reservoir	Vented 5 L food-grade bottle, highest shelf; outlet above settled sludge; S0 raw sample port.	Provides static head. Add a low-level float switch before unattended operation.
1 Solids	Top: washed coarse gravel. Middle: graded quartz sand. Bottom: support gravel and nylon retaining mesh.	Reduces particles before sorbents and UV. Wash until rinse water is clear; prevent media escape.
2 pH conditioning	High-calcium limestone cartridge with a hydraulic bypass.	Use for acidic mine drainage. Bypass if raw water is already neutral/alkaline; excessive pH can damage downstream performance.
3 Metal polish	Separate, replaceable natural-zeolite and characterized biochar layers/cartridges.	Potential polishing only. Metals removal is source-dependent and requires breakthrough testing plus lab confirmation.
4 Fluoride	Activated alumina baseline, or validated calcium-modified biochar/sorbent.	Do not claim a fixed capacity from literature. Competing ions, pH and contact time control breakthrough.
5 Sensor cell	Opaque flow-through cell; pH, TDS and turbidity probes; drain and calibration ports.	Place after all media and before UV. Exclude sunlight, bubbles and electrical cross-talk.
6 Barrier/outlet	Low-flow meter, validated stainless UV-C reactor, normally-closed tap solenoid.	No unmonitored treatment component between sensor cell and UV. Keep UV inaccessible and interlocked.

Scale warning

A nominal 2 L bottle may hold only about 1.2-1.5 L of packed media. At 0.25 L/min, empty-bed contact time is roughly 5-6 minutes; at 1 L/min it is only about 1-1.5 minutes. The YF-S201 is a poor match for the slower, more chemically credible bottle flow. For deployment, the proposed 3-4 inch diameter, 2 m columns provide a much larger bed volume, but their contact time and capacity still require pilot data.

Two source-specific operating modes

Mode	Recommended route	Release evidence
F: high-fluoride groundwater	Solids -> limestone bypassed unless needed -> fluoride sorbent -> polish -> sensor cell -> UV.	Fluoride breakthrough curve, pH/TDS/turbidity, metals panel, E. coli/coliform and validated UV dose.
M: acidic/mine-affected	Solids -> limestone -> oxidation/settling if needed -> zeolite/biochar -> fluoride cartridge only if fluoride is confirmed -> sensors -> UV.	Full dissolved/total metals panel, pH/alkalinity, sulfate, TDS, turbidity, microbial panel and UV validation.

4. Exact sensor placement

Device	Place it here	Installation detail	Why
pH probe	First position in final sensor cell, after all media and before UV.	Vertical or within manufacturer angle; glass bulb continuously wet; low-velocity bypass pocket; calibration access.	Controls limestone bypass/chemical acceptability and sees final treated chemistry.
TDS probe	Same cell downstream of pH, physically separated.	No bubbles; add temperature probe; isolate or time-multiplex excitation if probes interfere.	Tracks dissolved-ion change. It cannot identify fluoride or metals.
Turbidity	Last probe in opaque cell, immediately before flow meter/UV.	Shield from ambient light; use bubble trap; avoid deposits on optical windows; provide zero/standard port.	Interlocks UV when particles may shield microorganisms.
Flow meter	After sensor cell and before UV.	Straight pipe per sensor requirement; arrow with flow; calibrate at actual gravity head.	Dose and clog monitoring. Downstream placement measures water actually reaching UV.
Voltage	Inside dry IP65 control panel across fused 12 V bus.	Never contact water; calibrated divider; shared reference or isolation as designed.	Detects low battery/brownout and protects fail-closed operation.
Optional fluoride ISE	Post-filter side-stream after fluoride stage; sample then drains to waste.	High-impedance amplifier; add TISAB to sample only; standards bracket 0.1-20 mg/L; temperature compensated.	Provides direct fluoride evidence. Reagent must never enter product water.
Raw/sample ports	S0 raw, then post-stage ports after stages 1-4.	Capped, disinfectable fittings; collect without contaminating product path.	Makes media failure and breakthrough diagnosable instead of guessing.

Mechanical rules for reliable readings

- Put the sensing cell downstream of the last media stage. A sensor before treatment is useful for research, but it cannot interlock the product tap by itself.
- Keep the turbidity optical path continuously flooded and free of bubbles. Ambient light and air bubbles can look like unsafe turbidity.
- Do not place the probes in the raw reservoir: stagnant gradients and settled solids will make readings unrepresentative.
- Keep mains/valve wires away from high-impedance pH and fluoride signals; use shielded cable, strain relief and star grounding.
- Add two level switches for the next revision: source-water-present and backwash-water-ready. Without them, automatic backwash can run dry.

5. Detection levels and controller thresholds

A standard limit, a sensor detection/quantification level and a controller set-point are different. The limits below are grounded in the Jal Jeevan Mission/IS 10500 table and WHO guidance; the guard bands are conservative project choices.

Parameter	Reference value	Warn	Lock/action	Required capability and verdict
pH	6.5-8.5; no relaxation	<6.7 or >8.3	<6.5 or >8.5; immediate <5 or >10	Verified error <=0.1 pH. PH-4502C is acceptable only as a calibrated prototype input.
Turbidity	1 NTU acceptable; 5 NTU permissible	>=0.8 NTU	>=1.0 NTU; immediate >=5	Need low-range 0-10 NTU measurement with <=0.1 NTU resolution and validated uncertainty. TS-300B is trend-only.
TDS	500 mg/L acceptable; 2000 conditional	>500 mg/L	>=900 mg/L with listed board: instrument ceiling/unknown	Board range 0-1000 ppm and about +/-10% full scale cannot verify 2000. TDS is non-specific.
Fluoride	1.0 mg/L acceptable; 1.5 permissible/WHO	>=1.0 mg/L	>=1.4 mg/L; immediate >=2.0	Direct method LOQ <=0.1 mg/L over at least 0.1-20 mg/L. Not measured by listed sensors.
Flow	Process baseline	<80% clean-bed flow	Backwash if <60% for 30 s during a request	YF-S201 is normally about 1-30 L/min. Use a calibrated 0.3-6 L/min device if the rig is slower.
12 V bus	Engineering value	<11.2 V	<10.8 V	Calibrate divider and separately verify Mega brownout behavior under valve/SIM800L load.

TDS policy

500 mg/L is the acceptable value; 2000 mg/L is conditional when no alternative source exists. The firmware warns at 500 but locks near 900 because the listed sensor cannot measure the conditional limit. A public unit needs a validated wider-range conductivity instrument and an explicit local release policy.

Metals release criteria

Analyte	Acceptable / permissible (mg/L)	Project LOQ goal (mg/L)	Status
Arsenic	0.01 / 0.05	<=0.001	Laboratory only
Lead	0.01 / no relaxation	<=0.001	Laboratory only
Cadmium	0.003 / no relaxation	<=0.0003	Laboratory only
Total chromium	0.05 / no relaxation	<=0.005	Laboratory only
Nickel	0.02 / no relaxation	<=0.002	Laboratory only
Mercury	0.001 / no relaxation	<=0.0001	Laboratory only
Manganese	0.1 / 0.3	<=0.01	Laboratory only
Copper	0.05 / 1.5	<=0.005	Laboratory only
Iron	1.0 / no relaxation	<=0.1	Laboratory only

LOQ goals are project QA targets near one-tenth of the applicable limit, not statutory values.

6. Service flow and real gravity backwash

Normal service is top-to-bottom through each vented bottle. Backwash must reverse through the selected bed from bottom to top. A separate elevated clean-water tank supplies the reverse-flow head; dirty water is isolated from the product line and sent to waste.

State	V1 feed	V2 NC tap	V3 clean backwash	V4 waste	UV
Locked/idle	Closed	Closed	Closed	Closed	Off
Ready	Open	Closed	Closed	Closed	Off
UV warm-up	Open	Closed	Closed	Closed	On
Dispense	Open	Open	Closed	Closed	On
Isolate/settle	Closed	Closed	Closed	Closed	Off
Reverse backwash	Closed	Closed	Open	Open	Off

- Use non-return valves so untreated and backwash water cannot enter the safe-water line.
- Backwash one media cartridge at a time in the field design. The simple prototype manifold can demonstrate the sequence, but it does not prove full-bed cleaning.
- Do not backwash a fluoride/adsorption cartridge unless the manufacturer or media study supports it; backwash can redistribute exhausted media but cannot regenerate sorption capacity.
- Require a backwash-tank high-level switch and source-tank low-level switch before enabling unattended automatic flushing.
- After backwash, divert initial rinse water to waste and require fresh safe sensor samples before enabling the tap.

Trigger logic

Establish a clean-bed flow baseline at the lowest intended tank level. Warn below 80% of baseline. During a genuine dispense request, if flow remains below 60% for 30 seconds, isolate, reverse-flush, settle and return to READY only after safe readings. Never interpret zero flow while nobody is requesting water as a clog.

7. Safe Arduino Mega electrical design

Rail/device	Correct supply	Protection and wiring rule
12 V valves/UV input	Fused 12 V battery/adaptor bus sized for simultaneous loads.	Separate branch fuses; flyback diode on DC coils; never carry valve current through a breadboard or Mega pin.
Mega/sensors/relay logic	Regulated 5 V buck with current margin and low ripple.	Star ground, local decoupling, waterproof glands. Verify relay input polarity at reset.
SIM800L	Dedicated 4.0 V rail, capable of 2 A bursts.	At least 100 uF low-ESR close to module (470-1000 uF is commonly prudent), short wide conductors, logic-level shifting to Mega UART.
Analog probes	Clean regulated 5 V only if module permits.	Shielded/split routing from relays and antenna. Consider galvanic isolation or timed power sequencing for pH/TDS interference.
UV-C	Manufacturer-rated driver, switched by suitable MOSFET/relay.	Opaque closed chamber, door/cover interlock, current/intensity feedback. Never expose eyes or skin.

Arduino Mega pin map

Pin	Function	Connection
D2	Flow pulse interrupt	YF-S201 yellow or replacement sensor pulse
D3	Demand/authorization	Active-low pushbutton or isolated RFID authorization output
D4	Manual backwash	Active-low service button
D6	UV enable	Logic-level MOSFET/driver input; not the LED power path
D7	Local alarm	Buzzer driver
D8-D11	Four valves	Tap, feed, backwash, waste relay inputs
A0-A4	Analog sensing	pH, TDS, turbidity, supply voltage, optional fluoride amplifier
Serial1 D18/D19	SIM800L UART	Bidirectional logic-level shifting; D18 TX1, D19 RX1

Valve warning

Confirm the exact CR02/CR03 actuator wiring before connection. A spring-return or power-open two-wire valve can be controlled by one switched channel; a polarity-reversing or separate OPEN/CLOSE actuator needs an H-bridge or two interlocked relays. The firmware assumes each valve can be commanded open by energizing one channel and returns closed when de-energized.

8. Firmware architecture and fail-closed behavior

The supplied sketch targets Arduino Mega 2560, uses no third-party libraries and is split into configuration, measurement, sensor, policy, actuator and GSM modules. It avoids dynamic String allocation and long blocking delays.

State	Tap	Purpose
STARTUP_LOCKED	Closed	Wait for calibrated valid sensors and six consecutive safe samples.
READY	Closed	Feed path is available; awaits an active-low authorized demand.
UV_WARMUP	Closed	UV enabled for configured warm-up; cancellation returns to READY.
DISPENSING	Open	Only state that energizes the normally-closed drinking-water solenoid.
FAULT_LOCKED	Closed	Any invalid critical sensor or persistent unsafe result; local alarm and rate-limited GSM event.
BACKWASH_*	Closed	Isolate, reverse flush and settle; dispensing never resumes automatically.

- All calibration flags ship false. This is intentional: the prototype cannot dispense on placeholder coefficients.
- Three consecutive ordinary unsafe samples lock the unit; extreme pH/turbidity/fluoride locks immediately. Six safe samples plus released demand are required to recover.
- The watchdog resets the controller after a software stall. Relay outputs are driven to the de-energized safe state before serial/sensor startup.
- SIM800L SMS/call handling is a non-blocking state machine and is disabled until a real phone number, power rail and bench test are supplied.
- CSV telemetry exposes state, measurements, warnings and fault bitmasks for judging and validation logs.

Validation status

The source has been compiled with strict host-side Arduino stubs and the safety-policy unit tests pass. That catches syntax and deterministic policy errors, but it is not a substitute for compiling with the installed Arduino AVR core, flashing a real Mega, injecting sensor voltages, checking relay reset states and testing power interruptions.

Recommended next firmware increments

- Add source-low, backwash-ready and enclosure-open interlocks.
- Add UV current/intensity feedback; a commanded UV output is not proof of delivered dose.
- Store calibration coefficients and media service counters in EEPROM with version and CRC.
- Add an SD or FRAM event log so safety evidence does not depend on GSM coverage.
- Replace time-only backwash with per-cartridge differential pressure or validated flow recovery.

9. Calibration and validation protocol

Test	Minimum protocol	Acceptance decision
pH	Fresh pH 4.00, 7.00 and 10.00 buffers at known temperature; fit slope/intercept; independent check buffer before each trial day.	Error ≤ 0.1 pH across bracket; stable within declared response time.
Turbidity	0, 1, 5 and 20 NTU standards in final opaque cell, including bubbles and ambient-light challenge.	If uncertainty near 1 NTU cannot support the lock, label TS-300B trend-only and install a low-range meter.
TDS/conductivity	Certified standards bracketing expected water; record temperature; compare against reference meter.	Define matrix-specific error. Upgrade range before using any 2000 mg/L policy.
Flow	Five gravimetric timed collections at high/mid/low tank levels, clean bed and partly loaded bed.	Set pulses/L and clean-flow baseline from data; use a sensor whose specified range includes every test.
Fluoride	0.5, 1.0, 1.5, 5, 10 and 15 mg/L standards/samples; TISAB-conditioned; standards bracket sample; lab cross-check.	LOQ ≤ 0.1 mg/L; bias small enough to defend 1.4 mg/L interlock guard band.
Media breakthrough	Real source water, worst-case competing ions and target flow; sample every stage and cumulative bed volumes.	Replacement point set before effluent reaches conservative warning; report capacity with uncertainty.
Microbiology/UV	E. coli and total coliform, UV transmittance, validated dose at maximum flow, lamp aging and power variation.	Not detectable in 100 mL; delivered UV dose meets validated design, commonly at least 40 mJ/cm ² for a conservative prototype target.

Breakthrough study design

- Report influent chemistry, media mass/particle size, bed dimensions, flow, empty-bed contact time, pH, temperature and cumulative bed volumes.
- Plot effluent/influent concentration (C/C₀) against bed volumes for fluoride and each target metal. Do not use a single removal percentage as cartridge life.
- Use actual Dhanbad water. Published fluoride values vary across sites and studies; design around verified source sampling rather than one headline maximum.
- Challenge at the lowest and highest operating pH and with relevant sulfate, bicarbonate, phosphate and competing ions.
- Retest after backwash and after storage to identify desorption, channeling and microbial regrowth.

10. Failure-mode and safety review

Failure	Risk	Required response	Gap / addition
Power loss/reset	Untreated water dispensed	Normally-closed tap and de-energized safe relay state.	Bench-test reset transients and brownouts with every load switching.
Sensor unplugged/stuck	False safe reading	Range/plausibility failure locks tap; calibration flags default false.	Add stuck-value/rate-of-change diagnostics and periodic reference checks.
High turbidity	UV shielding	Lock at 1 NTU project set-point; divert to waste.	Use a validated low-range sensor and clean optical windows.
UV fails while commanded	Microbial exposure	Tap must close.	Add intensity/current feedback; present hardware list lacks this proof.
Source empty	False clog/backwash, pump/valve cycling	Block dispense/backwash and alert.	Add source low-level switch.
Backwash tank empty	Dry backwash and no cleaning	Abort and remain locked.	Add backwash-ready level switch.
Valve stuck/miswired	Cross-contamination	Hydraulic check valves and position verification.	Add limit feedback or flow-path proof for field unit.
GSM unavailable	Remote alert missed	Safety remains local and fail-closed; buzzer/log continues.	Never make water safety dependent on cellular service.
Media exhausted	Fluoride/metals breakthrough	Replace on measured breakthrough/service counter; lock if direct fluoride rises.	No sensor in current list proves metals removal; scheduled lab QA is mandatory.

Drinking-water release rule

Until the complete treatment train has passed chemical, metals, microbial and UV validation, label all output **NOT FOR DRINKING - TEST WATER ONLY**. Arduino interlocks reduce risk; they do not certify potability.

11. Demonstration and judging strategy

The strongest presentation is not a claim that inexpensive sensors solve every contaminant. It is a transparent safety architecture that knows its measurement limits and produces auditable evidence.

Demo moment	What judges see	Evidence shown
1. Gravity	Water moves down the labelled staircase with no service pump.	Measured raw-to-tap head and live flow at high/low tank levels.
2. Safe dispense	Authorized demand starts UV warm-up, then opens NC tap.	State and calibrated readings in serial dashboard/log.
3. Fault injection	Unplug sensor or introduce a safe simulated over-limit voltage; tap closes and alarm/SMS event is queued.	Fail-closed trace and fault bitmask; do not use hazardous contaminants on stage.
4. Clog/backwash	Restricted line causes persistent low flow; service isolates and reverse route opens to waste.	Before/after flow recovery; drinking path remains closed.
5. Science	Team displays fluoride breakthrough and metals lab reports rather than relying on TDS.	C/C0 curves, uncertainty, cartridge replacement point and source-specific treatment mode.

Differentiators worth defending

- Source-adaptive mode selection: limestone is used only when source chemistry needs it.
- Direct fluoride side-stream with reagent isolation, rather than a false TDS proxy.
- Hardware fail-closed tap, verified sensor health and UV interlock path.
- Replaceable cartridges with measured breakthrough curves and local service counters.
- Gravity-first hydraulics and local safety behavior that continue without cloud or GSM.
- Evidence dashboard that reports uncertainty and explicitly marks parameters as online, inferred or laboratory-only.

One-sentence pitch

ProtoWave is a gravity-first, source-adaptive treatment platform whose Arduino Mega controller refuses to dispense when measured quality, sensor health or the disinfection barrier is unsafe - while direct fluoride and laboratory metals evidence prevent low-cost sensors from making claims they cannot support.

12. Research basis and selected sources

1. Government of India, Jal Jeevan Mission - WQMS Framework / IS 10500 parameter table
2. WHO - Fluoride in Drinking-water
3. WHO - Water quality and health: review of turbidity
4. Jal Jeevan Mission - Technical Expert Committee report on IoT water monitoring
5. Arduino - Mega 2560 Rev3 official documentation
6. DFRobot - Analog TDS Sensor SEN0244 specifications and conversion
7. DFRobot - Analog pH Sensor SEN0161 specifications
8. DFRobot - Turbidity Sensor SEN0189 specifications
9. Sensors 2020 - Low-cost turbidity sensor evaluation including TS-300B
10. DATAQ - YF-S201 flow sensor specifications
11. SIMCom - SIM800L hardware design guide
12. NEMI - Standard Methods 4500-F C fluoride ion-selective electrode method
13. Shams et al. - Calcium-modified biochar for fluoride removal
14. US EPA - Activated alumina fluoride treatment design manual
15. Motsi et al. - Natural zeolite removal of metals from acid mine drainage
16. USGS - Limestone armoring in acid mine drainage systems
17. US EPA - Ultraviolet Disinfection Guidance Manual
18. Portable flow anodic-stripping voltammetry for As, Pb and Cd
19. Qualitative assessment of mine water in Western Jharia Coalfield
20. Groundwater fluoride dynamics and cited Dhanbad/Gharbar concentration context

Interpretation note

Literature removal efficiencies are not portable design capacities. Published Dhanbad-area source chemistry varies by site and season. The final media selection, service interval and release decision must be based on representative local samples, accredited analytical methods and a documented uncertainty budget.

DELIVERABLE STATUS

Exact stepped placement schematic, labelled concept render, Arduino Mega firmware, host smoke test, safety-policy unit tests and this technical design are included. Remaining work is physical calibration, Arduino AVR compilation/flash testing, hydraulic commissioning and accredited water-quality validation.